Year 13 Further Maths - Discrete random variables

Similar questions - Expectation, variance and linear combinations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The discrete random variable X has the following distribution, where a, b and c are constants.

x	0	1	2	3
$\mathrm{P}(X=x)$	a	b	c	0.15

It is given that $\mathrm{E}(X)=1.45$ and $\mathrm{Var}(X)=0.9475$.

(a) Determine the values of a, b and c. **[5]**

(b) The random variable Y is defined by $Y=5-3X$. Find $\mathrm{Var}(Y)$. **[1]**

(c) Fifty independent observations of X are made; T is the number for which $X=3$. Find $\mathrm{Var}(T)$. **[2]**

2 The discrete random variable D has the following probability distribution, where a is a constant.

d	1	3	5	7
$\mathrm{P}(D=d)$	0.1	a	0.3	0.2

Determine the value of $\mathrm{Var}(4D-3)$. **[7]**

3 In this question you must show detailed reasoning. The discrete random variable V has the probability distribution below, where p and q are constants.

v	0	1	2	3
$\mathrm{P}(V=v)$	p	q	0.2	0.1

It is given that $\mathrm{E}(V)=\mathrm{Var}(V)$. Determine the values of p and q. **[8]**

4 The random variable X is equally likely to take any of the n integer values from $m+1$ to $m+n$ inclusive. It is given that $\mathrm{E}(2X)=39$ and $\mathrm{Var}(2X)=21$. Determine the values of m and n. **[7]**

5 The discrete random variable W has the distribution $\mathrm{U}(11)$. The independent discrete random variable V has the distribution $\mathrm{U}(5)$.

(a) For constants m and n with $m>0$, $\mathrm{E}(mW+nV)=0$ and $\mathrm{Var}(mW+nV)=1$. Determine the exact values of m and n. **[5]**

(b) The random variable T is the mean of three independent observations of W. Explain whether the Central Limit Theorem can be used to say that T is approximately normally distributed. **[1]**

Question 1
(a) $a+b+c=0.85$; $\mathrm{E}(X)$: $b+2c+0.45=1.45$ so $b+2c=1$. $\mathrm{E}(X^2)=\mathrm{Var}+\mathrm{E}^2=0.9475+2.1025=3.05=b+4c+1.35$, so $b+4c=1.7$. Hence $c=0.35$, $b=0.3$, $a=0.2$
(b) $\mathrm{Var}(5-3X)=9\times 0.9475=8.5275$
(c) $T\sim \mathrm{B}(50,0.15)$, $\mathrm{Var}(T)=50(0.15)(0.85)=6.375$

Question 2
$a=1-0.6=0.4$. $\mathrm{E}(D)=0.1+1.2+1.5+1.4=4.2$; $\mathrm{E}(D^2)=0.1+3.6+7.5+9.8=21$; $\mathrm{Var}(D)=21-17.64=3.36$. $\mathrm{Var}(4D-3)=16\times 3.36=53.76$

Question 3
$p+q=0.7$. $\mathrm{E}(V)=q+0.4+0.3=q+0.7$. $\mathrm{E}(V^2)=q+0.8+0.9=q+1.7$. $\mathrm{Var}=\mathrm{E}(V^2)-\mathrm{E}(V)^2=q+1.7-(q+0.7)^2$. Setting equal to $q+0.7$: $(q+0.7)^2=1$, so $q+0.7=1$ ($\mathrm{E}(V)>0$), $q=0.3$ and $p=0.4$

Question 4
$\mathrm{E}(X)=m+\dfrac{n+1}{2}=19.5$ and $\mathrm{Var}(X)=\dfrac{n^2-1}{12}=\dfrac{21}{4}$, so $n^2-1=63$, $n=8$, $m+4.5=19.5$, $m=15$

Question 5
(a) $\mathrm{E}(W)=6$, $\mathrm{Var}(W)=\frac{120}{12}=10$; $\mathrm{E}(V)=3$, $\mathrm{Var}(V)=\frac{24}{12}=2$. $6m+3n=0\Rightarrow n=-2m$. $\mathrm{Var}=10m^2+2n^2=18m^2=1$, so $m=\dfrac{1}{3\sqrt{2}}=\dfrac{\sqrt{2}}{6}$, $n=-\dfrac{\sqrt{2}}{3}$
(b) No: T is the mean of only three observations, which is too small a sample for the CLT to give a reliable approximation (although the parent distribution is symmetric, it is not normal)